

ODA-GS: Occlusion- and Distortion-aware Gaussian Splatting for Indoor Scene Reconstruction

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Abstract

In this work, we aim to address the quality degradation issues of indoor scene reconstruction using 3D Gaussian Splatting (3DGS). Existing methods enhance reconstruction quality by exploiting learned geometric priors like Signed Distance Functions (SDF), but these come with significant computational costs. We analyze the traditional 3DGS training process and identify key factors contributing to quality degradation: over-reconstruction and gradient dilution during the densification stage, and the occurrence of distorted/redundant Gaussians during the post-optimization stage. To tackle these issues, we introduce ODA-GS, a novel framework that modifies 3DGS with tailored modules. During densification, we employ occlusion-aware gradient accumulation to prevent gradient dilution and use homo-directional gradients to mitigate over-reconstruction. In the post-optimization stage, we introduce post-pruning to eliminate distorted and redundant Gaussians, thereby enhancing visual quality and reducing computational overhead. Tested on the ScanNet++ and Replica datasets, ODA-GS outperforms several baselines both qualitatively and quantitatively.

CCS Concepts

• **Computing methodologies** → **Machine learning**; **Computer graphics**; **Point-based models**.

*Both authors contributed equally to the paper

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1 Introduction and Related Work

3D reconstruction has been a subject of study for decades. Recently, 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023] has marked a significant breakthrough in 3D reconstruction due to its realistic quality and real-time rendering capabilities. Given multi-view images and structure points estimated by Structure-from-Motion (SfM), 3DGS employs 3D Gaussians as explicit scene representations optimized for pose, scale, opacity, and anisotropic color. The typical training process of 3DGS consists of two stages: *densification*

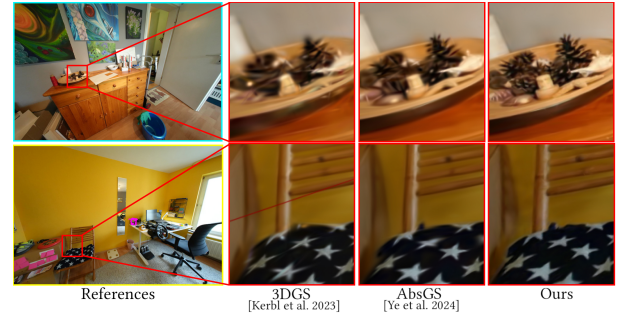


Figure 1: Visual comparisons with previous methods demonstrate that ODA-GS outperforms in both detail reconstruction (top row) and the reduction of high-frequency artifacts (bottom row).

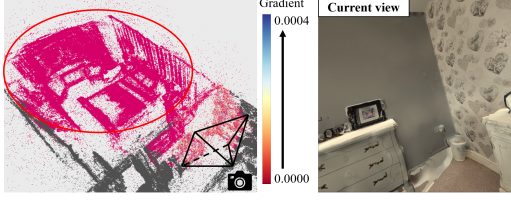


Figure 2: Illustration of gradient dilution: Gaussians in the view frustum (right) are colored by their gradients (left). Occluded Gaussians (red circle) have small gradients but still contribute to accumulation, causing gradient dilution.

and *post-optimization*. In the densification stage, the focus is on optimizing and identifying Gaussians via a gradient accumulation mechanism, where 3DGS records the positional gradients of visible Gaussians and computes the average gradient across all views. These average gradients then guide the splitting, cloning, or pruning of Gaussians, which is crucial as it determines the growth or reduction of 3D Gaussians. After 15,000 iterations, the densification process ceases, and the post-optimization stage begins to fine-tune the parameters of the Gaussians.

In indoor scene reconstruction, some research adopts learned geometric priors, such as Signed Distance Functions (SDF), during the training process to enhance reconstruction quality [Xiang et al. 2024; Yu et al. 2024]. However, these approaches require training additional deep learning models, significantly increasing overall computational complexity. In contrast, we have identified key factors in the training process of 3DGS that directly contribute to the quality degradation of indoor scene reconstruction. Based on these insights, we propose tailor-made strategies to effectively address these issues. The key factors are elaborated as follows.

Over-reconstruction. The densification process has the issue of over-reconstruction, where 3DGS represents intricate regions with a few large Gaussians, resulting in blurred reconstruction. Some methods address this issue by modifying gradient accumulation [Ye et al. 2024; Zhang et al. 2024a] or introducing loss in the frequency domain [Zhang et al. 2024b]. However, these methods still overlook geometric factors, limiting their effectiveness in indoor scenes.

Gradient dilution. In indoor scenes, occlusions caused by walls and other structures significantly impact the densification process. As shown in Fig. 2, Gaussians in occluded regions (e.g., behind the walls) have small positional gradients, while those in non-occluded areas have larger gradients. Due to the simple viewing frustum

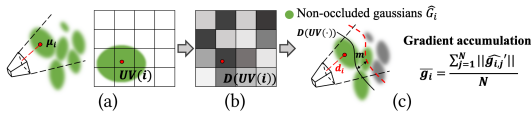


Figure 3: Occlusion-aware gradient accumulation. We project the Gaussian center μ_i into pixel space to get its UV coordinate $UV(i)$ (a). We then sample the depth value $D(UV(i))$ (b). Finally, we perform an occlusion test to identify non-occluded Gaussians and accumulate their gradient (c).

culling, both occluded and non-occluded Gaussians are given equal weight in the gradient accumulation. Consequently, the gradients from occluded Gaussians dilute the overall gradient, limiting the densification process's ability to perform splitting and cloning operations. This dilution leads to inevitable quality degradation.

Distorted and redundant Gaussians. During the post-optimization stage, we identify two phenomena. First, due to inaccurate camera pose estimation or inconsistent lighting across views, some Gaussians become distorted to minimize reconstruction loss. Second, the number of Gaussians with low opacity continues to increase, leading to computational redundancy. While previous methods effectively address redundant Gaussians by assessing their importance and applying quantization methods to compress the scene [Fan et al. 2023; Papantonakis et al. 2024], they do not address distorted Gaussians, which significantly impact visual quality.

In this work, we introduce ODA-GS, a novel framework that carefully modifies the training process of 3DGS with tailor-made modules to address the aforementioned issues. Fig. 1 illustrates the effectiveness of our approach, while Fig. 4 provides an overview of ODA-GS. In the densification stage, we alleviate the gradient dilution by employing the *occlusion-aware gradient accumulation* to accumulate gradients only for non-occluded Gaussians. Additionally, we adopt the homo-directional gradient [Ye et al. 2024] to mitigate over-reconstruction. During the post-optimization stage, we introduce *post-pruning* to eliminate distorted and redundant Gaussians, improving visual quality and reducing computational overhead. We test ODA-GS on the ScanNet++ [Yeshwanth et al. 2023] and Replica [Straub et al. 2019] datasets. The experimental results indicate that our method outperforms several baselines both qualitatively and quantitatively. In summary, we make the following contributions:

- We introduce ODA-GS, a novel framework that modifies the training process of 3DGS to address key issues in indoor scene reconstruction.
- Identify key factors in the 3DGS training process that lead to quality degradation in indoor scene reconstruction, including the over-reconstruction, the gradient dilution, and the distorted/redundant Gaussians.
- Propose a novel occlusion-aware gradient accumulation technique to mitigate gradient dilution.
- Introduce a post-pruning technique to eliminate distorted and redundant Gaussians during the post-optimization stage.

2 Methodology

2.1 Occlusion-Aware Gradient Accumulation

We propose a novel occlusion-aware gradient accumulation strategy to address gradient dilution and refine the details of the indoor scenes. We conduct an occlusion test on each visible Gaussian and then accumulate gradients only for the non-occluded Gaussians, as illustrated in Fig. 3.

We first render the depth map, D [Chung et al. 2023]. For each Gaussian, we project its center, μ_i , into the pixel space to obtain its UV coordinates, $UV(i)$, and look up the depth value, $D(UV(i))$, from D using these coordinates. Next, we apply the occlusion test on each visible Gaussian by comparing the depth of each Gaussian

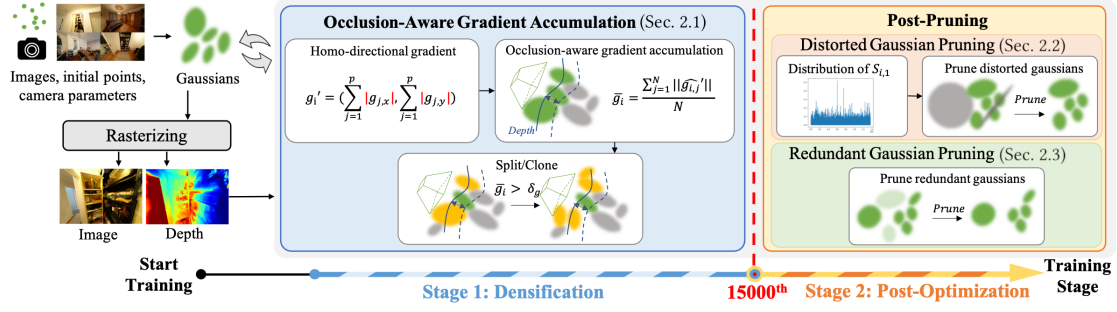


Figure 4: ODA-GS overview. In the densification stage, we employ the occlusion-aware gradient accumulation (Sec. 2.1) to alleviate the gradient dilution. In the post-optimization stage, we perform the post-pruning to eliminate the distorted Gaussians (Sec. 2.2) and the redundant Gaussians (Sec. 2.3) to reduce the high-frequency artifacts and improve the performance.

from the camera, d_i , with $D(UV(i))$. To account for the potential discrepancies between the rendered and true depth, we add a small margin, m , to the sampled depth, $D(UV(i))$, to avoid the overly strict occlusion constraints. We denote the non-occluded Gaussians as $\hat{G}_i = \{G_i \mid d_i < D(UV(i) + m)\}$. We perform gradient accumulation on these non-occluded Gaussians. For each non-occluded gaussian, $\hat{G}_i$, we accumulate their homo-directional gradient, $\hat{g}'_{i,j}$ [Ye et al. 2024], across the views, $\{v_j \mid j = 1, \dots, N\}$. We express the average gradient, $\bar{g}_i$, as

$$\bar{g}_i = \frac{\sum_{j=1}^N \|\hat{g}'_{i,j}\|}{N}, \quad (1)$$

where i denotes the index of Gaussian. The average gradient, $\bar{g}_i$, is then used to guide the splitting and cloning of Gaussians. Consequently, it improves the overall accuracy and detail of 3D scene reconstructions.

2.2 Distorted Gaussian Pruning

Here, we detail our distorted Gaussian pruning method that reduces high-frequency artifacts during post-optimization. We analyze the scale $S_i \in \mathbb{R}^3$ of each Gaussian, represented as $S_i = [S_{i,1}, S_{i,2}, S_{i,3}]$, where $S_{i,1}$, $S_{i,2}$ and $S_{i,3}$ represents the axis with the scale from large to small. We then classify the distorted Gaussians into needle-like Gaussians and large Gaussians that have a stretched appearance from certain viewpoints. To estimate how much a Gaussian distort, we define its distortion ratio according to the largest scale $S_{i,1}$ against the other two smaller scales $S_{i,2}$ and $S_{i,3}$ as $R_{i,2} = \frac{S_{i,1}}{S_{i,2}}$ and $R_{i,3} = \frac{S_{i,1}}{S_{i,3}}$. Needle-like Gaussians, G_N , then are characterized by judging the distortion ratios, $R_{i,2}$ and $R_{i,3}$, with an outlier threshold, δ_S . We set $\delta_S = \mu + 3\sigma$, where μ and σ are the mean and standard deviation of $S_{i,1}$ at iteration 15,000. Large Gaussians, G_L , have a larger scale than needle-like Gaussians. Thus, we detect them with a higher outlier threshold. For each Gaussian, G_i , we denote the above distorted Gaussians, G_N and G_L , as

$$G_N = \{G_i \mid R_{i,2} > \delta_N, R_{i,3} > \delta_N, \text{ and } S_{i,1} > \delta_S\}, \text{ and} \\ G_L = \{G_i \mid S_{i,1} > 10 \cdot \delta_S\},$$

where the distortion threshold δ_N is set to 30. We then detect and prune these distorted Gaussians to reduce the high-frequency artifacts.

2.3 Redundant Gaussian Pruning

Lastly, we apply the redundant Gaussian pruning to reduce the number of Gaussians without compromising reconstruction quality. During post-optimization, many low-opacity Gaussians contribute minimally to the rendered image yet consume the computational resources. To address this, we periodically prune Gaussians with opacity below a threshold δ_o , set to 0.005.

3 Experiments

Experimental settings. We conduct the experiments on an NVIDIA V100 GPU (32GB), using the same loss function as 3DGS [Kerbl et al. 2023]. We apply our method on 10 scenes selected from ScanNet++ [Yeshwanth et al. 2023] and Replica [Straub et al. 2019] datasets with each scene comprising around 1,000 images and testing every 8-th image. All scenes are trained for 100 epochs. We report the metrics of the evaluation including PSNR, SSIM, and LPIPS. We also compare our method with 3DGS, GaussianPro [Cheng et al. 2024], PixelGS [Zhang et al. 2024a], and AbsGS [Ye et al. 2024]. Note GaussianPro [Cheng et al. 2024] is compared only with ScanNet++ due to its need for ordered data.

Quantitative and qualitative results. In this experiment, we compare the results of our method with the previous works on the ScanNet++ and Replica datasets, as shown in Table 1. Our method achieves the best performance in terms of PSNR, SSIM, and LPIPS for both datasets. Fig. 5 shows the visual comparisons, where our occlusion-aware gradient accumulation effectively addresses gradient dilution and enhances the scene details like the small objects and the fine textures (see Fig. 5, top row). Additionally, the distorted Gaussian pruning reduces high-frequency artifacts and leads to a better visual quality (see Fig. 5, bottom row).

Ablation study. In this experiment, we validate the performance of each step of our method with the following settings: without occlusion-aware gradient accumulation but with homo-directional gradient (Ours-w/o OCC), without distorted Gaussian pruning (Ours-w/o DP), without redundant Gaussian pruning (Ours-w/o RP), and with all the modules (Ours-full). The quantitative results, presented in Table 2, demonstrate that our method with all the modules achieves the best performance on the ScanNet++ and

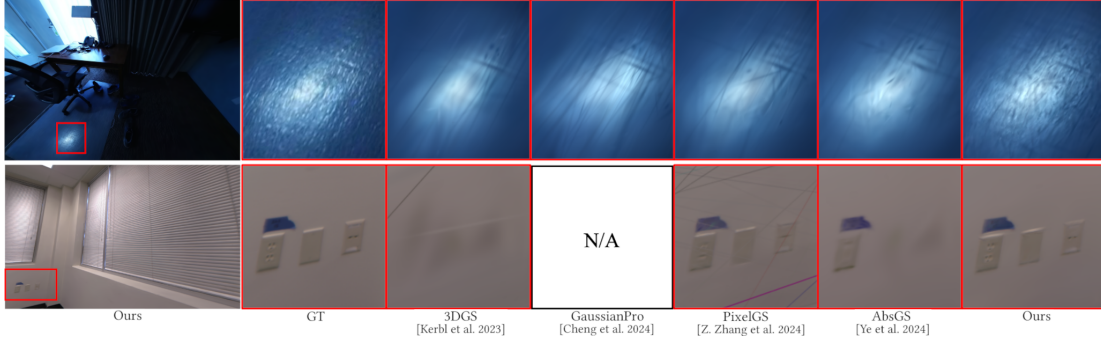


Figure 5: Qualitative comparison on ScanNet++ (top row) and Replica (bottom row) datasets. GaussianPro is compared only with ScanNet++ due to its need for ordered data.

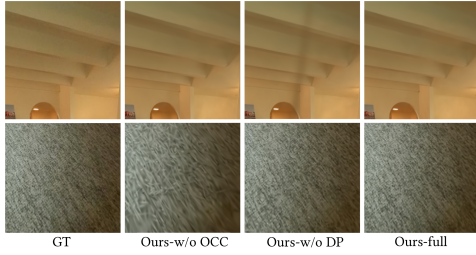


Figure 6: Qualitative comparison of ablation study.

Replica datasets. Additionally, reducing redundant Gaussians preserves quality while further decreasing training time and model size by around 20%. Fig. 6 shows that the incorporating occlusion-aware gradient accumulation improves the reconstruction of fine details, while the distorted Gaussian pruning effectively removes high-frequency artifacts.

4 Conclusion

In this work, we introduce ODA-GS, a novel framework that addresses key issues in 3DGS for indoor scene reconstruction. By identifying and tackling factors such as over-reconstruction, gradient dilution, and distorted/redundant Gaussians, ODA-GS significantly enhances reconstruction quality. Our tailored strategies include occlusion-aware gradient accumulation and homo-directional gradients during the densification stage, as well as post-pruning in the post-optimization stage. Experimental results on ScanNet++ and Replica datasets demonstrate that ODA-GS outperforms several baselines both qualitatively and quantitatively, offering a robust solution for high-quality indoor scene reconstruction.

Table 1: Quantitative comparison on ScanNet++ and Replica.

Method	ScanNet++ [Yeshwanth et al. 2023]			Replica [Straub et al. 2019]		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
3DGS [Kerbl et al. 2023]	31.37	0.9361	0.1773	33.47	0.9555	0.1287
GaussianPro [Cheng et al. 2024]	31.93	0.9387	0.1795	-	-	-
PixelGS [Zhang et al. 2024a]	32.18	0.9400	0.1749	33.75	0.9584	0.1231
AbsGS [Ye et al. 2024]	32.50	0.9436	0.1683	35.63	0.9741	0.0954
Ours	33.07	0.9494	0.1464	35.85	0.9747	0.0850

Table 2: Quantitative comparison of ablation study.

Method	ScanNet++ [Yeshwanth et al. 2023]					Replica [Straub et al. 2019]				
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Time $\downarrow$	Size $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Time $\downarrow$	Size $\downarrow$
Ours-w/o OCC	32.69	0.9456	0.1617	-	-	35.61	0.9733	0.0955	-	-
Ours-w/o DP	32.99	0.9490	0.1477	-	-	35.50	0.9740	0.0856	-	-
Ours-w/o RP	33.02	0.9494	0.1464	2h36m	631.71MB	35.69	0.9745	0.0852	1h45m	807MB
Ours-full	33.04	0.9494	0.1465	2h28m	529.86MB	35.85	0.9747	0.0850	1h34m	533.67MB

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